BUSINESS OPERATIONS EXCELLENCE · GROUP 3

From 10 to 12 Excavators a Day

Applying Operations Excellence to the machine shop at Volvo CE India, Peenya, Bengaluru



Industrial visit: 8 September 2026

Presentation: 3–4 November 2026 · BBA, Group 3

Peenya Industrial Area, Bengaluru

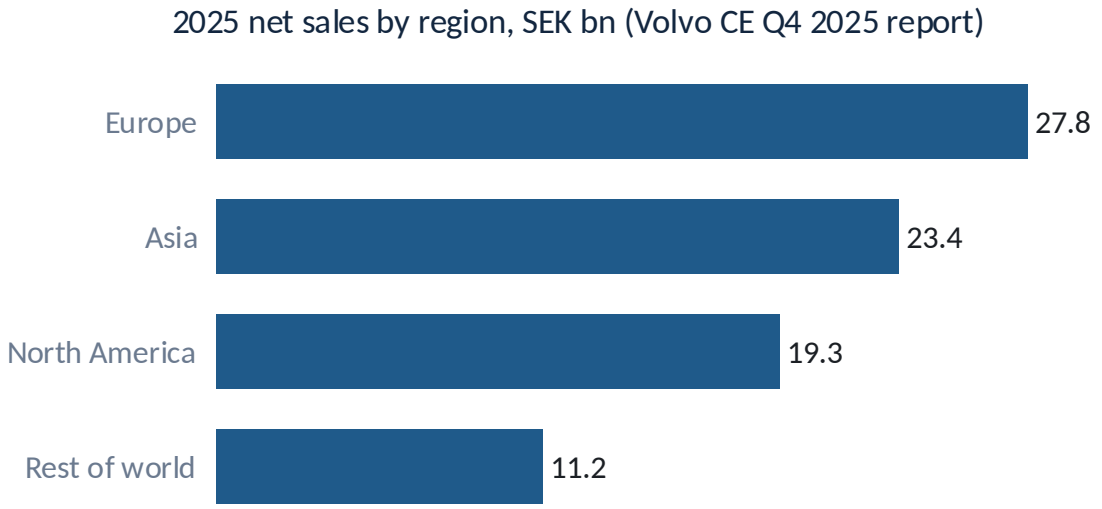
Volvo CE worldwide at a glance

SEK 81.6 bn
Net sales 2025 (≈ ₹ 74,000 cr)

13.3%
Adjusted operating margin 2025

~13,000
Employees, 85 nationalities

180
Markets served via dealers



Brands VOLVO · premium machines | ROKBAK · haulers, Scotland

Plants in Sweden, Germany, France, Poland, UK, USA, Brazil, South Korea, China and India

SDLG stake (70%) sold in Sep 2025; dealer Swecon bought in Jan 2026

1832 — Munktell starts the business in Eskilstuna, Sweden

1950 — Volvo buys Bolinder-Munktell

1995 — Renamed Volvo Construction Equipment

1997 — Volvo CE India starts in Bengaluru

2025 — Exits SDLG; new Eskilstuna excavator plant

2026 — Buys Swecon; EC650 launched in India

Global portfolio: application & tonnage



Electric models offered (e.g., ECR25, EC230, L25, L120 Electric). Bars show approx. operating weight; haulers show rated payload (Volvo CE spec sheets).

Volvo CE India: the Peenya hub

Where
India HQ and plant in Peenya Industrial Area, Bengaluru

History
Site since 1978; Volvo CE India since 1997; ₹90 cr upgrade and new excavator line in 2011

What it builds
Crawler excavators, soil compactors and asphalt pavers; EC210 is ~70% localised

Market reach
19 dealers, 180+ touchpoints; ~6.5% Indian CE share (2024); exports to emerging markets

Engineering
Bengaluru Technology Centre: R&D and localisation for India and the world

From steel to machine at Peenya

1 Receiving — Steel plate, castings, purchased parts

2 Fabrication — Cutting and welding of booms, arms, frames

3 Machine shop — **Boring, milling, drilling: our focus**

4 Paint shop — Frames primed and painted

5 Assembly — Line paced by the digital takt board

6 Test & dispatch — Function test, final inspection, dispatch

Recent milestones

Dec 2025 · EXCON, Bengaluru

EC215 “Made in Bharat” 20–22 t excavator unveiled

Sep 2026 · bauma CONEXPO INDIA

EC650, a 65 t mining excavator, launched

2027 · Next

Localised medium wheel loaders after SDLG exit

India offering: application & tonnage

Segment	Models	Tonnage	Typical applications in India
20–22 t crawler excavators	EC210, EC215, EC220 ● **Built at Peenya**	20–22 t	Highways, irrigation canals, urban infra, stone quarries
Heavy crawler excavators	EC300, EC380, EC480	30–48 t	Quarrying, mass earthmoving, overburden removal
Mining excavators	EC650 (new, Sep 2026), EC750	65–75 t	Coal and iron-ore mines
Wheel loaders	Premium L-series incl. L120 Electric	≈20 t +	Mining, cement, steel plants, ports (~100 a year)
Soil compactors	SD110 / SD110C ● **Built at Peenya**	≈11 t	Highway embankments, sub-grade compaction
Asphalt compactors	DD90C, DD100C, PT220	≈9–10 t	Bituminous surfacing of roads
Asphalt pavers (ABG)	P4370D wheeled; P5320D–P8820D tracked ● **Built at Peenya**	Paving width	National highways and expressways

WHY +20% MATTERS

≈70%

of India's 30,000–35,000 excavators a year are in the 20 t class, the core of the Peenya line

4% → 8%+

Volvo CE India's 20 t share in 2024 and its goal to more than double it

More demand at Peenya means the machine shop must deliver more units per shift.

The brief: 10 → 12 excavators a shift

10
excavators per 8-hour shift today (assumed)

12
target per shift: +20%, i.e. 2 more units

420
minutes available per shift after breaks

42 → 35
takt time, minutes per excavator

Where the 420 minutes come from

480	–	30	–	20	–	10	=	420
shift		lunch		2 tea breaks		start-up & 5S		available

Takt = available time ÷ demand
420 ÷ 10 = 42 min today → 420 ÷ 12 = 35 min target

Ground rules we set ourselves

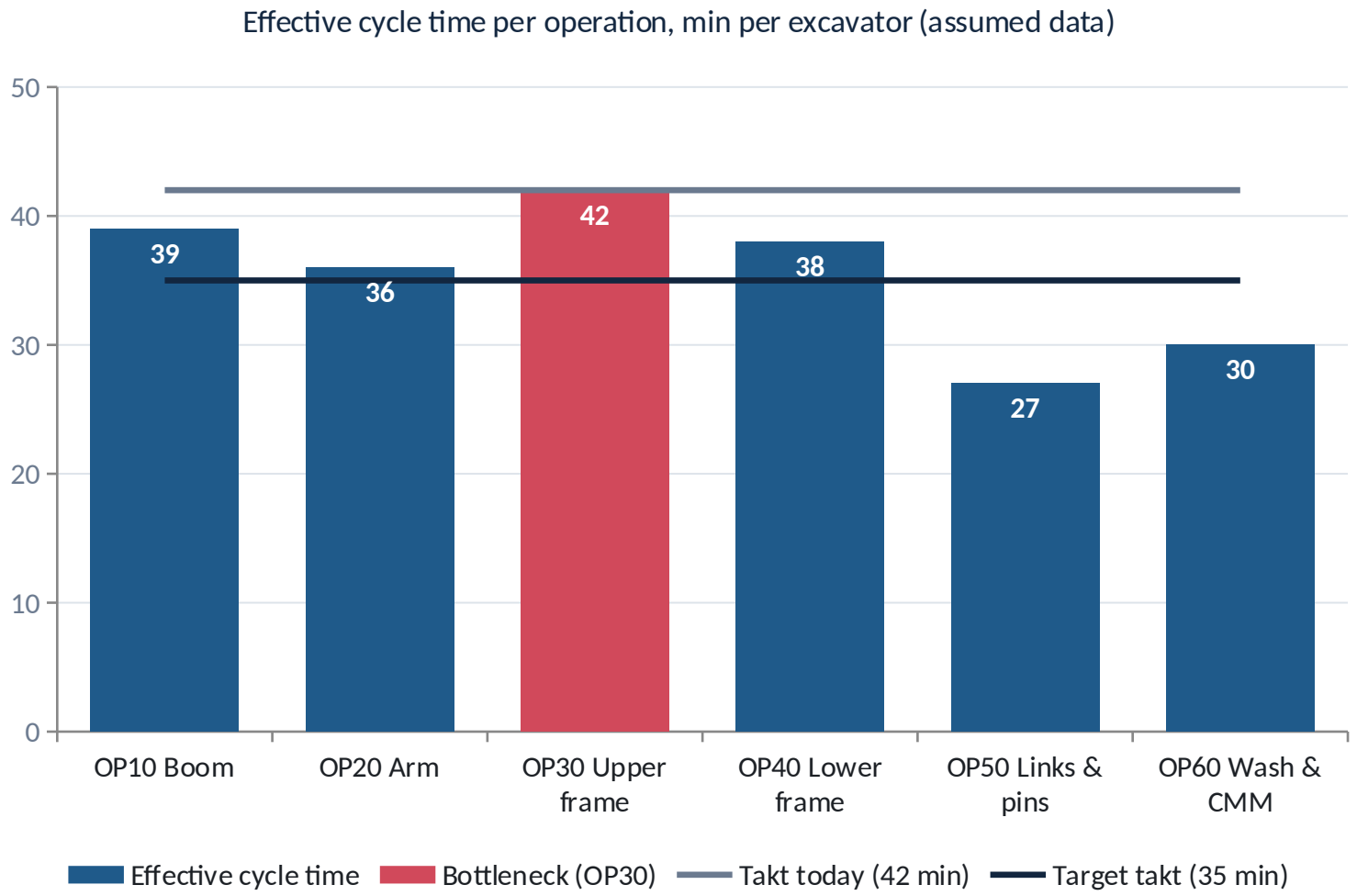
- Same 8-hour shift and the same headcount
- No compromise on quality (first-pass yield, Cpk) or safety
- Minimal capex: fix the process before buying capacity
- Scope: machine shop feeding the EC210/215/220 line

Overtime or a 2nd shift
Adds cost; fatigue raises the quality risk

Buy another machining centre
₹4–5 cr capex and a 9–12 month wait

Operations Excellence ✓
Low capex, ~12 weeks, builds lasting capability

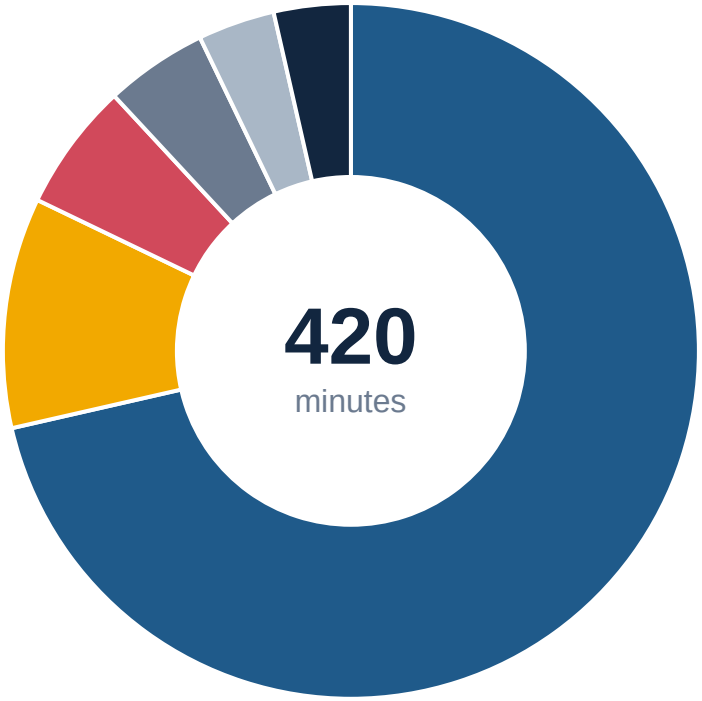
Gemba finding: OP30 sets the pace



Gemba findings (assumed)

1. OP30 (upper frame, 5-face machining centre) runs at 42 min = takt: zero spare capacity
2. 4 of 6 operations are slower than the 35 min target takt
3. Frames wait for the shared overhead crane and first-off QC sign-off
4. Manual dial-gauge alignment and 3 model changeovers a day (EC210/215/220)

Where OP30's 420 minutes go today



- **Value-adding machining** 300 min
  10 good frames × 30 min each
- **Changeovers** 45 min
  Availability loss
- **Breakdowns & minor stops** 25 min
  Availability loss
- **Waiting: crane, parts, QC** 20 min
  Performance loss
- **Slow set-up: dial-gauge alignment** 15 min
  Performance loss
- **Rework / re-machining** 15 min
  Quality loss

Availability
(420 − 45 − 25) ÷ 420
83.3%

Performance
28-min ideal × 10.5 frames ÷ 350 min
84.0%

Quality
10 good ÷ 10.5 frames machined
95.2%

OEE = A × P × Q
World-class benchmark ≈ 85%
66.7%

120 lost minutes a day = 4 excavators' worth of OP30 time. Win back about 60 of them and the shop makes 2 more machines with no new machine and no rushing.

Levers 1–3: stabilise and feed OP30

Theory of Constraints: fix the bottleneck (OP30) first, because a minute saved there is a minute of plant output; then roll out to OP10, OP20 and OP40.

VPS · PROCESS STABILITY

1 · SMED quick changeover

45 → 15 min/day

+1.0 excavator of capacity

- Preset the next frame's fixture and tools while the current frame is machined
- Zero-point quick clamps and a tool presetter: 15 → 5 min per change
- Sequence frames by family (EC210/215/220)

KPI minutes per changeover (goal ≤ 5)

VPS · PROCESS STABILITY

2 · TPM for the machines

25 → 12 min/day

+0.4 excavator of capacity

- Operators do a 10-min daily clean-lubricate-inspect-tighten check
- Planned maintenance calendar and spindle vibration monitoring
- Critical spares kit at the machine to cut repair time

KPI MTBF up, MTTR down, breakdown min/shift

VPS · JUST-IN-TIME

3 · JIT flow and kitting

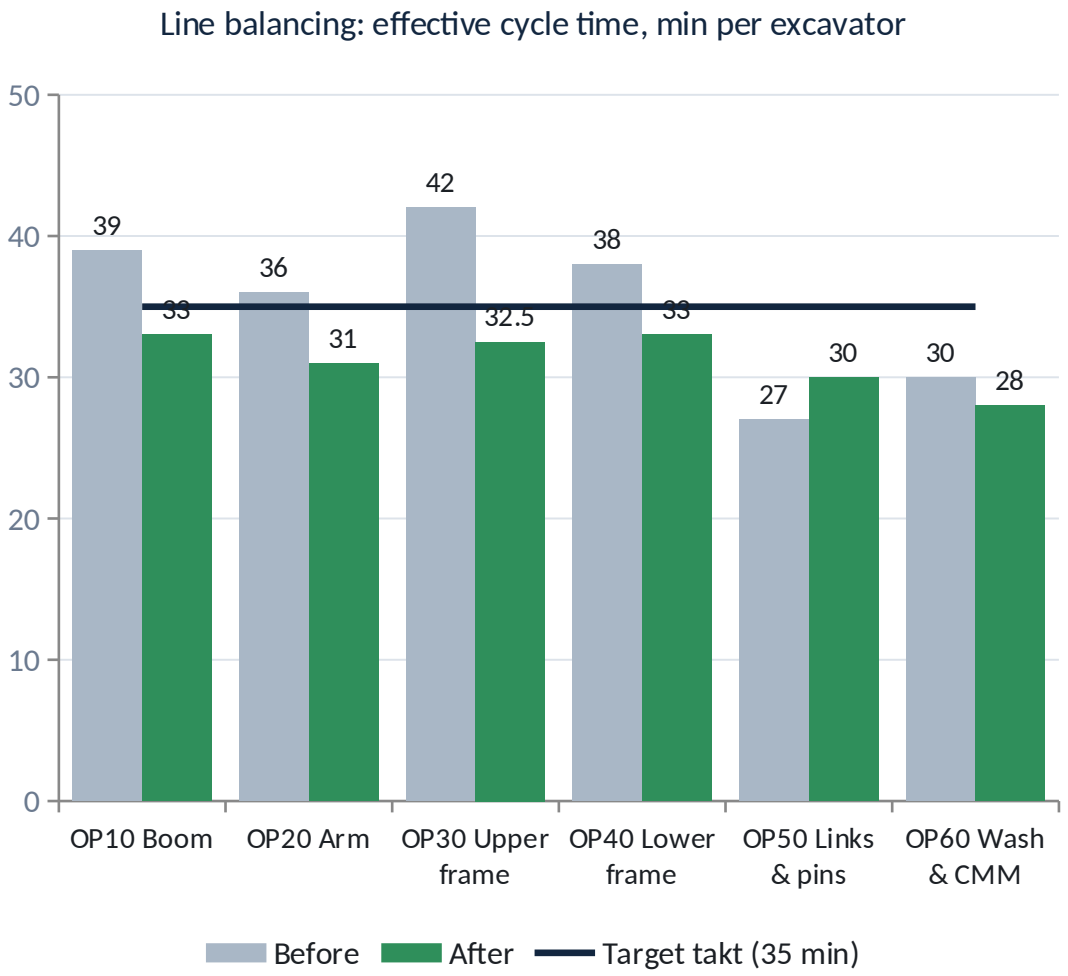
20 → 6 min/day

+0.5 excavator of capacity

- FIFO lane with a 2-frame buffer so OP30 is never starved
- Kanban pull of weldments from fabrication; tool kits per model
- Dedicated jib crane at OP30: no waiting for the shared crane

KPI minutes OP30 waits for a frame (goal ≤ 6)

Levers 4–6: standardise, protect quality



4 · Standard work & line balancing

Slow set-up 15 → 6 min; cycle 30 → 29 min · +0.7 excavator

- Time-study standard work; drilling moved from OP10/40 to OP50
- Probing replaces dial-gauge set-up; CNC paths optimised

5 · Built-in quality: poka-yoke, SPC, jidoka

Rework 15 → 6 min · +0.3 excavator

- Poka-yoke locators: a frame cannot be clamped the wrong way
- SPC on pin bores (Cpk ≥ 1.33); machine stops on tool wear

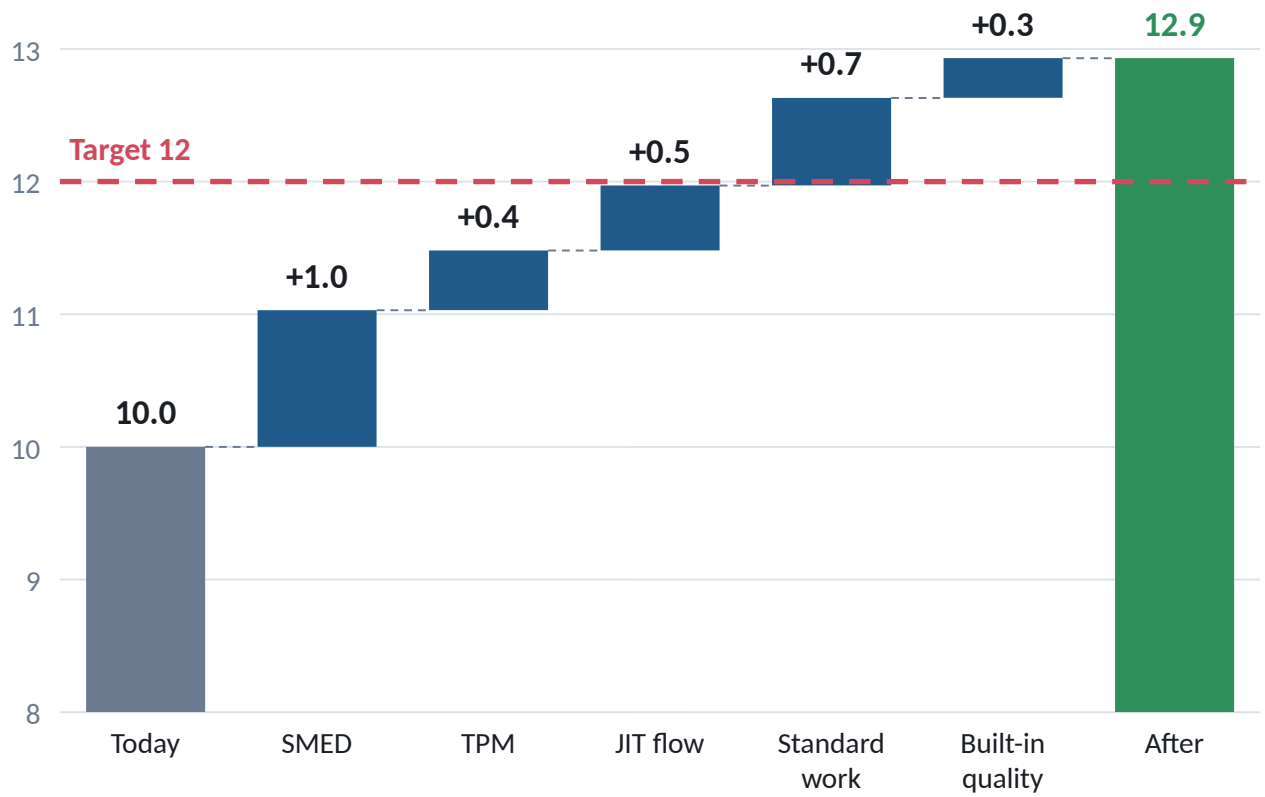
6 · 5S, visual management & kaizen

Sustains the gains shift after shift

- 5S, shadow boards; hourly output and OEE on the digital board
- Daily tier meetings, PDCA kaizen, operator skill matrix

Result: capacity for 12.9, plan for 12

OP30 capacity bridge, excavators per shift



Axis starts at 8 so the increments are visible. Values use the assumed data model.

KPI	Today	After 12 weeks
Output, excavators per shift	10	**12 (capacity 12.9)**
OEE at OP30	66.7%	**86% capability**
Effective cycle time at OP30	42 min	**32.5 min**
Changeover per model	15 min	**5 min**
Breakdown loss per shift	25 min	**12 min**
First-pass yield at OP30	95.2%	**98.4%**

Robust: even if only 70% of the gains hold, capacity is 12.05, still above target.

Upside not counted: relief operator runs OP30 through tea breaks (+20 min ≈ +0.7 unit).

12-week roadmap to 12 a day

WEEKS 1–2 · PLAN

Diagnose

- Gemba walks, time study
- OEE baseline at OP30
- Kaizen team formed

Output **10.0 / shift**

›

WEEKS 3–6 · DO

Quick wins

- 5S and visual boards
- SMED workshop at OP30
- FIFO lane, kits, jib crane

Output **11.0 / shift**

›

WEEKS 7–10 · DO

Structural fixes

- TPM steps 1–3
- Probing, quick clamps, CNC
- Poka-yoke; move drilling

Output **11.6 / shift**

›

WEEKS 11–12 · CHECK–ACT

Lock in

- Verify FPY and Cpk
- Freeze standard work
- Roll out to OP10/20/40

Output **12.0 / shift**

Waste, not speed

Most of the 2 extra machines come from winning back 75 of OP30's 120 lost minutes

Quality by design

No inspection step removed; validated feeds; FPY < 98% or Cpk < 1.33 means stop and fix

VPS sustains it

Standard work, daily tier meetings and kaizen keep OP30 under the 35-min takt

Thank you. Questions welcome.